

Apache Kafka 4.3.1 — KRaft Setup Guide

Complete commands for macOS and Windows: generate UUID → format storage → start Kafka → create topic → producer → consumer → verify messages.

Important

Kafka 4.3.1 uses KRaft and does not use ZooKeeper. The UUID and storage-format commands are normally required only for the initial setup of a fresh Kafka data directory. After setup, normally start the Kafka server directly.

Part A — macOS

1. Go to the Kafka folder

```
cd ~/kafka_2.13-4.3.1
```

2. Generate the KRaft UUID

```
bin/kafka-storage.sh random-uuid
```

Copy the UUID returned by this command.

3. Format KRaft storage

```
bin/kafka-storage.sh format --standalone -t YOUR-UUID -c config/server.properties
```

Replace YOUR-UUID with the UUID from step 2.

4. Start the Kafka server

```
bin/kafka-server-start.sh config/server.properties
```

Keep this terminal running.

5. Open a new Terminal and go to Kafka

```
cd ~/kafka_2.13-4.3.1
```

6. Create a topic

```
bin/kafka-topics.sh --create --topic test --bootstrap-server localhost:9092
```

7. Verify the topic

```
bin/kafka-topics.sh --list --bootstrap-server localhost:9092
```

8. Start the producer in another Terminal

```
cd ~/kafka_2.13-4.3.1
bin/kafka-console-producer.sh --topic test --bootstrap-server localhost:9092
```

Then type messages such as:

```
Hello Kafka
This is my first message
Kafka is working
```

9. Start the consumer in another Terminal

```
cd ~/kafka_2.13-4.3.1
bin/kafka-console-consumer.sh --topic test --bootstrap-server localhost:9092
```

10. Consumer should display new messages

```
Hello Kafka
This is my first message
Kafka is working
```

11. Read messages from the beginning

```
bin/kafka-console-consumer.sh --topic test --from-beginning --bootstrap-server localhost:9092
```

Part B — Windows

Assume Kafka is installed at C:\kafka\kafka_2.13-4.3.1. If your folder is elsewhere, replace the path.

1. Go to the Kafka folder

```
cd C:\kafka\kafka_2.13-4.3.1
```

2. Generate the KRaft UUID

```
bin\windows\kafka-storage.bat random-uuid
```

Copy the UUID returned by this command.

3. Format KRaft storage

```
bin\windows\kafka-storage.bat format --standalone -t YOUR-UUID -c config\server.properties
```

Replace YOUR-UUID with the UUID from step 2.

4. Start the Kafka server

```
bin\windows\kafka-server-start.bat config\server.properties
```

Keep this Command Prompt running.

5. Open a new Command Prompt

```
cd C:\kafka\kafka_2.13-4.3.1
```

6. Create a topic

```
bin\windows\kafka-topics.bat --create --topic test --bootstrap-server localhost:9092
```

7. Verify the topic

```
bin\windows\kafka-topics.bat --list --bootstrap-server localhost:9092
```

8. Start the producer in another Command Prompt

```
cd C:\kafka\kafka_2.13-4.3.1
bin\windows\kafka-console-producer.bat --topic test --bootstrap-server localhost:9092
```

Then type messages such as:

```
Hello Kafka
This is my first message
Kafka is working
```

9. Start the consumer in another Command Prompt

```
cd C:\kafka\kafka_2.13-4.3.1
bin\windows\kafka-console-consumer.bat --topic test --bootstrap-server localhost:9092
```

10. Consumer should display new messages

```
Hello Kafka
This is my first message
Kafka is working
```

11. Read messages from the beginning

```
bin\windows\kafka-console-consumer.bat --topic test --from-beginning --bootstrap-server localhost:9092
```

Quick Reference — macOS vs Windows

Task	macOS/Linux	Windows
Generate UUID	bin/kafka-storage.sh random-uuid	bin\windows\kafka-storage.bat random-uuid
Format storage	bin/kafka-storage.sh format ...	bin\windows\kafka-storage.bat format ...
Start server	bin/kafka-server-start.sh config/server.properties	bin\windows\kafka-server-start.bat config\server.properties
Create topic	bin/kafka-topics.sh --create ...	bin\windows\kafka-topics.bat --create ...
Producer	bin/kafka-console-producer.sh ...	bin\windows\kafka-console-producer.bat ...
Consumer	bin/kafka-console-consumer.sh ...	bin\windows\kafka-console-consumer.bat ...

Normal startup after initial formatting: start the Kafka server only. Do not run the storage format command again against an already initialized data directory unless you intentionally want to reinitialize it.